Mathematical Models for Computer Systems Engineering
Machine Problem 2
Group 4
Members: Bilanes, Marinas, Nicolas, Ong

$$y' + \frac{1}{x}y = x \quad or \quad y' = x - \frac{y}{x} \qquad x > 0, \qquad y(3) = 5$$

Task 1: *Analytical Solution*

1. Express the given as Mdx + Ndy = 0

$$y' + \frac{1}{x}y = x \Rightarrow dy + (\frac{1}{x}y - x)dx = 0$$

So:

$M(x, y) = \frac{1}{x}y - x$

$N(x, y) = 1$

2. Test if the Differential Equation is Exact

$$M_y = \frac{1}{x} \quad , \quad N_x = 0$$

Thus: Not Exact

3. If not an Exact D.E., determine the Integrating Factor *u.*

Case 1: $u = u(x)$

$$\frac{M_y - N_x}{N} = \frac{\frac{1}{x} - 0}{1} = \frac{1}{x}$$

So:

$$u(x) = e^{\int \frac{M_y - N_x}{N} dx} = e^{\int 1/x} = e^{\ln x} = x$$

4. Test the new Differential Equation for Exactness

$$xM\,dx \;+\; xN\,dy \;=\; 0 \;\Rightarrow x(\tfrac{1}{x}y \;-\; x)dx \;+\; xdy \;=\; 0$$

Solving it:

$$(y \;-\; x^2)dx \;+\; xdy \;=\; 0$$

Exactness Test:

$$M_y = 1 \quad , \quad N_x = 1$$

Thus: It is now Exact

5. Test the new Differential Equation using Analytical Method (should solve using the assigned Method for solving Exact D.E.'s from Lecture 9.1)

Method 2: Working with $\int Ndy$

$$\int Ndy \;=\; \int x\,dy \;=\; xy$$

Using the formula:

$$\frac{\partial}{\partial x}[\int Ndy] \;+\; h'(x) \;=\; M$$

So:

$$\frac{\partial}{\partial x}[xy] \;+\; h'(x) \;=\; y \;-\; x^2$$

$$y \;+\; h'(x) \;= y \;-\; x^2 \Rightarrow h'(x) \;=\; -x^2$$

Then Integrate $h'(x)$:

$$h(x) \;=\; -\frac{x^3}{3}$$

Thus:

$$F(x,y) \;=\; xy - \frac{x^3}{3} \;=\; C$$

6. Using the given initial conditions, determine the arbitrary constant C of the solution.

Using the condition, $y(3) = 5$

$$(3)(5) - \frac{3^3}{3} = C \Rightarrow 15 - 9 = 6$$
$$C = 6$$

7. Express the solution in explicit form, i.e. y = f(x)

$$xy - \frac{x^3}{3} = 6 \Rightarrow y = \frac{x^2}{3} + \frac{6}{x}$$

Thus:

$$y_{true}(x) = \frac{x^2}{3} + \frac{6}{x} \qquad , x > 0$$

Using this equation to tabulate the Analytical Method (to compare it to the Numerical Methods):

x	y_{true}
3	5.0000
4	6.8333
5	9.5333
6	13.0000
7	17.1905

Task 1.1: *Numerical Solution*

1. **Euler's Method**

Formula/s to be used:

$$y_{n+1} = y_n + f(x_n, y_n)h \quad , h = 1$$

$x_0 = 3, y_0 = 5 \Rightarrow Given$

Getting (y_1)

$y_1 = y_0 + f(x_0, y_0)h \Rightarrow 5 + f(3,5)(1)$

$f(3,5) = 3 - \frac{5}{3} = 1.3333$

$y_1 = 5 + 1.3333 = 6.3333$

Getting (y_2)

$y_2 = y_1 + f(x_1, y_1)h \Rightarrow 6.3333 + f(4, 6.3333)(1)$

$f(4, 6.3333) = 4 - \frac{6.3333}{4} = 2.4167$

$y_2 = 6.3333 + 2.4167 = 8.7500$

Getting (y_3)

$y_3 = y_2 + f(x_2, y_2)h \Rightarrow 8.7500 + f(5, 8.7500)(1)$

$f(5, 8.7500) = 5 - \frac{8.7500}{5} = 3.2500$

$y_3 = 8.7500 + 3.2500 = 12.0000$

Getting (y_4)

$$y_4 = y_3 + f(x_3, y_3)h \Rightarrow 12 + f(6, 12)(1)$$

$$f(6, 12) = 6 - \frac{12}{6} = 4.0000$$

$$y_4 = 12.0000 + 4.0000 = 16.0000$$

Thus:

x	y_{true}	y_{Euler}	$\lvert\varepsilon_t\rvert$ %
3	5.0000	5.0000	0
4	6.8333	6.3333	7.32%
5	9.5333	8.7500	8.22%
6	13.0000	12.0000	7.69%
7	17.1905	16.0000	6.93%

2. 4th Order Runge-Kutta Method

Formulas to be used:

$$k_1 = f(x_n, y_n)$$

$$k_2 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_1)$$

$$k_3 = f(x_n + \frac{h}{2}, y_n + \frac{h}{2}k_2)$$

$$k_4 = f(x_n + h, y_n + hk_3)$$

$$y_{n+1} = y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4) \quad , \quad h = 1$$

$x_0 = 3,\ y_0 = 5 \Rightarrow Given$

Getting (y_1)

$k_1 = f(3\ ,\ 5) = 1.33333333$

$k_2 = f(3 + \frac{1}{2}\ ,\ 5 + \frac{1}{2}(1.33333333)) = f(3.5,\ 5.66666667)$

$= 3.5 - \frac{5.66666667}{3.5} = 1.88095238$

$k_3 = f(3 + \frac{1}{2}\ ,\ 5 + \frac{1}{2}(1.88095238)) = f(3.5,\ 5.94047619)$

$= 3.5 - \frac{5.94047619}{3.5} = 1.80272109$

$k_4 = f(3 + 1\ ,\ 5 + 1.80272109) = f(4,\ 6.80272109)$

$= 4 - \frac{6.80272109}{4} = 2.29931973$

$y_1 = 5 + \frac{1}{6}(1.33333333 + 2(1.88095238) + 2(1.80272109) + 2.29931973\ )$

$= 6.83333333$

Getting (y_2)

$k_1 = f(4\ ,\ 6.83333333) = 4 - \frac{6.83333333}{4} = 2.29166667$

$k_2 = f(4 + \frac{1}{2}\ ,\ 6.83333333 + \frac{1}{2}(2.29166667)) = f(4.5,\ 7.97916667)$

$= 4.5 - \frac{7.97916667}{4.5} = 2.72685185$

$k_3 = f(4 + \frac{1}{2}\ ,\ 6.83333333 + \frac{1}{2}(2.72685185)) = f(4.5, 8.19675926)$

$= 4.5 - \frac{8.19675926}{4.5} = 2.67849794$

$k_4 = f(4 + 1\ ,\ 6.83333333 + 2.67849794) = f(5,\ 9.51183127)$

$= 5 - \frac{9.51183127}{5} = 3.09763375$

$y_2 = 6.83333333 + \frac{1}{6}(2.29166667 + 2(2.72685185) + 2(2.67849794) + 3.09763375\ )$

$= 9.53333333$

Getting (y_3)

$k_1 = f(5\ ,\ 9.53333333) = 5 - \frac{9.53333333}{5} = 3.09333333$

$k_2 = f(5 + \frac{1}{2}\ ,\ 9.53333333 + \frac{1}{2}(3.09333333)) = f(5.5,\ 11.0800)$

$$= 5.5 - \frac{11.0800}{5.5} = 3.48545455$$

$$k_3 = f(5 + \frac{1}{2}, 9.53333333 + \frac{1}{2}(3.48545455)) = f(5.5, 11.27606061)$$

$$= 5.5 - \frac{11.27606061}{5.5} = 3.44980716$$

$$k_4 = f(5 + 1, 9.53333333 + 3.44980716) = f(6, 12.98314049)$$

$$= 6 - \frac{12.98314049}{6} = 3.83614325$$

$$y_3 = 9.53333333 + \frac{1}{6}(3.09333333 + 2(3.48545455) + 2(3.44980716) + 3.83614325)$$

$$= 13.0000$$

Getting (y_4)

$$k_1 = f(6, 13) = 6 - \frac{13}{6} = 3.83333333$$

$$k_2 = f(6 + \frac{1}{2}, 13 + \frac{1}{2}(3.83333333)) = f(6.5, 14.91666667)$$

$$= 6.5 - \frac{14.91666667}{6.5} = 4.20512820$$

$$k_3 = f(6 + \frac{1}{2}, 13 + \frac{1}{2}(4.20512820)) = f(6.5, 15.10256410)$$

$$= 6.5 - \frac{15.10256410}{6.5} = 4.17652860$$

$$k_4 = f(6 + 1, 13 + 4.17652860) = f(7, 17.17652860)$$

$$= 7 - \frac{17.17652860}{7} = 4.54621020$$

$$y_3 = 13 + \frac{1}{6}(3.83333333 + 2(4.20512820) + 2(4.17652860) + 4.54621020)$$

$$= 17.19047619$$

Thus:

x	y_{true}	y_{RK4}	$\lvert\varepsilon_t\rvert$ %
3	5.0000	5.0000	0
4	6.8333	6.8333	0
5	9.5333	9.5333	0
6	13.0000	13.0000	0
7	17.1905	17.1905	0

Task 2: *MATLAB Script*

For the following user input statements, the group used while loops and if-else statements to continuously prompt the user until valid values are entered for all five inputs. These include:

- choice of numerical method
- x0 (lower limit)
- xf (upper limit)
- y0 (value of y at x0)
- h (interval/step size)

Validate User Input for Method Selection

```
C/C++
disp('Choose Numerical Method to be used:');
disp('  1) Euler''s Method');
disp('  2) Runge-Kutta (Order 4)');
choice = input('What is your choice? ');
i = 0;
while i == 0
   choice = input('What is your choice? ');
   if choice == 1
       disp('You have chosen Euler''s Method!');
       i = 1;
   elseif choice == 2
       disp('You have chosen Runge-Kutta (Order 4)!');
       i = 1;
   else
       disp('Invalid choice. Please enter 1 or 2.');
   end
end
```

Defining the RHS f(x,y) and later the exact solution from the IC

```
C/C++
f = @(x,y) x - (1./x).*y;   % domain x>0
```

This defines the RHS of your ODE as a MATLAB anonymous (inline) function.

Validate User Input for Values x0, xf, y0, & h

```
C/C++
x0 = 0;
while ~(isscalar(x0) && isfinite(x0) && x0>0)
    disp (' ');
    x0 = input('Enter the LOWER limit for x range (>0): ', 's');
    x0 = str2double(x0);
    if isnan(x0)
        disp('Invalid input. Please enter a numeric value.');
        x0 = 0;

    elseif ~(isscalar(x0) && isfinite(x0) && x0>0)
        disp('Invalid, x0 must be finite and > 0.');
    end
end

% Upper limit for x range
xf = x0; % xf can't equal x0
while ~(isscalar(xf) && isfinite(xf) && xf>x0)
    disp (' ');
    xf = input('Enter the UPPER limit for x range (> lower): ', 's');
    xf = str2double(xf);
    if isnan(xf)
        disp('Invalid input. Please enter a numeric value.');
        xf = x0;
    elseif ~(isscalar(xf) && isfinite(xf) && xf>x0)
        disp('Invalid, xf must be finite and > x0.');
    end
end

% y value at given x value
y0 = 1/0;
while ~(isscalar(y0) && isfinite(y0))
    disp (' ');
    y0 = input(['Enter the value for y at x = ' num2str(x0) ': '], 's');
    y0 = str2double(y0);

    if isnan(y0)
        disp('Invalid input. Please enter a numeric value.');
        y0 = 1/0;
    elseif ~(isscalar(y0) && isfinite(y0))
        disp('Invalid, y0 must be finite.');
    end

end
```

```
% Step size
h = 0;

while ~(isscalar(h) && isfinite(h) && h>0)
    disp (' ');
    h  = input('Enter the interval or step size (>0): ', 's');
    h = str2double(h);
    if isnan(h)
        disp('Invalid input. Please enter a numeric value.');
        h = 0;
    elseif ~(isscalar(h) && isfinite(h) && h>0)
        disp('Invalid, h must be finite and > 0.');
    end
end
```

Next, the group generates the range of x-values starting from x0 to xf using interval h and checks for errors like xf - x0 not being a multiple of h and any x value being equal or close to zero. The part also initializes y and then solves the differential equation using either Euler's Method or Runge-Kutta (4th order) to compute the successive y values. Afterward, it calculates the exact solution using the initial condition and determines the percent relative error between numerical and exact results.

Generation of Range

```
C/C++
% Generate range
x = x0:h:xf;
if abs(x(end)-xf) > 1e-12
    fprintf('Note: (xf - x0) is not a multiple of h. Using x_end_adj =
%.12g.\n', x(end));
end
if any(abs(x) < eps)
    error('Grid hits x=0; choose x0>0 and steps that avoid 0.');
end

n = numel(x);
y = zeros(1,n); y(1) = y0;
```

Numerical Solutions

```
C/C++
% Numerical solution
if choice == 1
    % Euler
    for i = 1:n-1
        y(i+1) = y(i) + h * f(x(i), y(i));
    end
else
    % RK4
    for i = 1:n-1
        k1 = f(x(i), y(i));
        k2 = f(x(i) + h/2, y(i) + (h/2)*k1);
        k3 = f(x(i) + h/2, y(i) + (h/2)*k2);
        k4 = f(x(i) + h,   y(i) + h*k3);
        y(i+1) = y(i) + (h/6)*(k1 + 2*k2 + 2*k3 + k4);
    end
end
```

Calculation of Exact Solution and Percent Relative Error

```
C/C++
% Exact solution that matches the entered IC
C = x0*(y0 - x0^2/3);
y_true = @(x_) x_.^2/3 + C./x_;
y_exact = y_true(x);

% Percent relative error
err = 100*abs((y_exact - y)./y_exact);
err(~isfinite(err)) = NaN;
```

Lastly, the results are collated into a tabular form, and a graph plotting the true and numerical solutions is done.

Tabulate Output

```
C/C++
% Table of Values
data = [x; y_exact; y; err]; % Combine data into one matrix

disp(' ');
fprintf('Tabulated solution:\n\n');

% Headers
fprintf('%5s%15s%15s%15s\n', 'x', 'y_true', method, '|εₓ| (%)');

% Line
fprintf('%5s%15s%15s%15s\n',repmat('-', 1, 5), repmat('-', 1, 15), repmat('-',
1, 15), repmat('-', 1, 15));

% Values
fprintf('%5.2f%15.5f%15.5f%15.2f\n', data);
```

Plotting the Graph

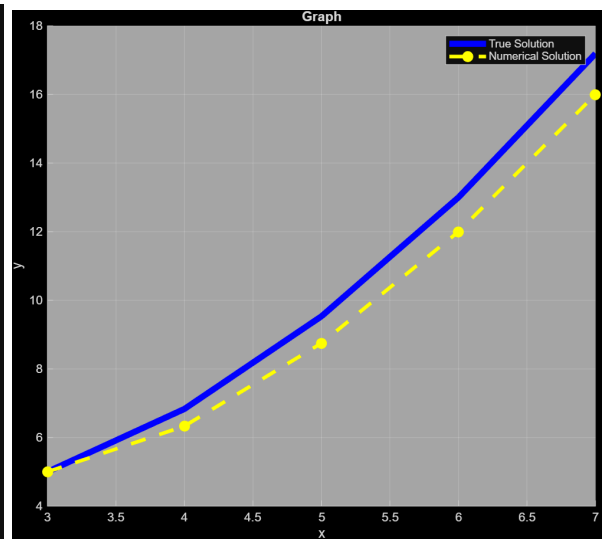
```
C/C++
% Plotting of Graph
disp(' ');

figure;
hold on;
set(gca, 'Color', [0.65 0.65 0.65 ]); % background color for graphing area

plot(x, y_exact, 'b', 'LineWidth', 5);
plot(x, y, 'y--o', 'MarkerFaceColor', 'y', 'LineWidth', 3);
legend('True Solution', 'Numerical Solution');
grid on;

hold off;

xlabel('x');
ylabel('y');
title('Graph');
```

Testing:

1. Using Euler's Method (Test 1)

Inputs:
choice (method specified) = 1 (Euler's Method)
x0 (lower limit) = 3
xf (upper limit) = 7
y0 (value of y at x0) = 5
h (interval/step size) = 1

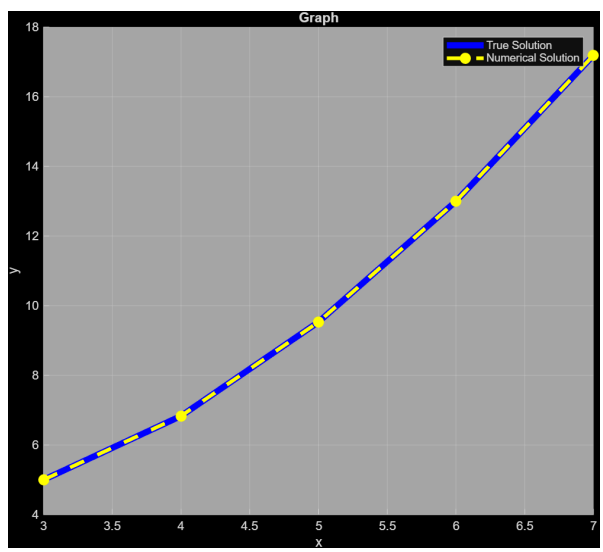
```
This script solves the differential equation dy/dx + (1/x)y = x,  x > 0

Choose Numerical Method to be used:
  1) Euler's Method
  2) Runge-Kutta (Order 4)
What is your choice? 1

You have chosen Euler's Method!

Enter the LOWER limit for x range (>0): 3

Enter the UPPER limit for x range (> lower): 7

Enter the value for y at x = 3: 5

Enter the interval or step size (>0): 1

Tabulated solution:

    x         y_true        y_Euler       |εₓ| (%)
-----------------------------------------------------
 3.00        5.00000        5.00000           0.00
 4.00        6.83333        6.33333           7.32
 5.00        9.53333        8.75000           8.22
 6.00       13.00000       12.00000           7.69
 7.00       17.19048       16.00000           6.93
```



2. Using RK4 Method (Test 1)

Inputs:
choice (method specified) = 2 (RK4 Method)
x0 (lower limit) = 3
xf (upper limit) = 7
y0 (value of y at x0) = 5
h (interval/step size) = 1

```
This script solves the differential equation dy/dx + (1/x)y = x,  x > 0

Choose Numerical Method to be used:
  1) Euler's Method
  2) Runge-Kutta (Order 4)
What is your choice? 2

You have chosen Runge-Kutta (Order 4)!

Enter the LOWER limit for x range (>0): 3

Enter the UPPER limit for x range (> lower): 7

Enter the value for y at x = 3: 5

Enter the interval or step size (>0): 1

Tabulated solution:

    x         y_true          y_RK4       |εₓ| (%)
-----------------------------------------------------
 3.00        5.00000        5.00000           0.00
 4.00        6.83333        6.83333           0.00
 5.00        9.53333        9.53333           0.00
 6.00       13.00000       13.00000           0.00
 7.00       17.19048       17.19048           0.00
```



3. Using Euler's Method (Test 2)

Inputs:

choice (method specified) = 1 (Euler's Method)

x0 (lower limit) = 0.5

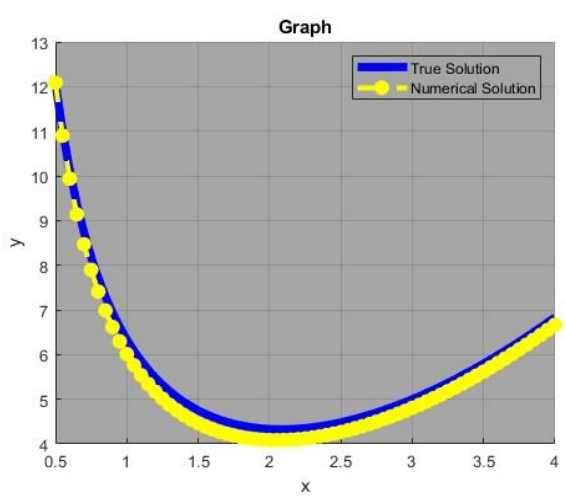
xf (upper limit) = 4

y0 (value of y at x0) = 12.08334

h (interval/step size) = 0.05

```
Choose Numerical Method to be used:
  1) Euler's Method
  2) Runge-Kutta (Order 4)
What is your choice? 1

You have chosen Euler's Method!

Enter the LOWER limit for x range (>0): 0.5

Enter the UPPER limit for x range (> lower): 4

Enter the value for y at x = 0.5: 12.08334

Enter the interval or step size (>0): 0.05

Tabulated solution:
```

x	y_true	y_Euler	$\lvert\varepsilon_t\rvert$ (%)
0.50	12.08334	12.08334	0.00
0.55	11.00993	10.90001	1.00
0.60	10.12001	9.93660	1.81
0.65	9.37161	9.13855	2.49
0.70	8.73477	8.46808	3.05
0.75	8.18750	7.89822	3.53
0.80	7.71334	7.40917	3.94
0.85	7.29966	6.98610	4.30
0.90	6.93667	6.61765	4.60
0.95	6.61663	6.29500	4.86
1.00	6.33334	6.01119	5.09
1.05	6.08179	5.76063	5.28
1.10	5.85788	5.53881	5.45
1.15	5.65823	5.34205	5.59
1.20	5.48000	5.16729	5.71
1.25	5.32084	5.01198	5.80
1.30	5.17872	4.87400	5.88
1.35	5.05195	4.75154	5.95
1.40	4.93905	4.64306	5.99
1.45	4.83877	4.54723	6.02
1.50	4.75000	4.46293	6.04
1.55	4.67180	4.38917	6.05
1.60	4.60334	4.32508	6.04
1.65	4.54387	4.26992	6.03
1.70	4.49275	4.22303	6.00
1.75	4.44941	4.18383	5.97
1.80	4.41334	4.15179	5.93
1.85	4.38408	4.12646	5.88
1.90	4.36123	4.10743	5.82
1.95	4.34442	4.09434	5.76
2.00	4.33333	4.08686	5.69
2.05	4.32766	4.08469	5.61
2.10	4.32714	4.08756	5.54
2.15	4.33153	4.09524	5.46
2.20	4.34061	4.10750	5.37
2.25	4.35417	4.12415	5.28
2.30	4.37203	4.14500	5.19
2.35	4.39403	4.16989	5.10
2.40	4.42000	4.19867	5.01
2.45	4.44981	4.23120	4.91
2.50	4.48333	4.26735	4.82
2.55	4.52044	4.30700	4.72
2.60	4.56103	4.35005	4.63
2.65	4.60499	4.39640	4.53
2.70	4.65222	4.44594	4.43
2.75	4.70265	4.49861	4.34
2.80	4.75619	4.55432	4.24
2.85	4.81276	4.61299	4.15
2.90	4.87230	4.67456	4.06
2.95	4.93473	4.73897	3.97
3.00	5.00000	4.80615	3.88
3.05	5.06805	4.87604	3.79
3.10	5.13882	4.94861	3.70
3.15	5.21226	5.02379	3.62
3.20	5.28833	5.10155	3.53
3.25	5.36699	5.18184	3.45
3.30	5.44818	5.26462	3.37
3.35	5.53188	5.34985	3.29
3.40	5.61804	5.43750	3.21
3.45	5.70663	5.52754	3.14
3.50	5.79762	5.61993	3.06
3.55	5.89098	5.71464	2.99
3.60	5.98667	5.81166	2.92
3.65	6.08467	5.91094	2.86
3.70	6.18496	6.01247	2.79
3.75	6.28750	6.11622	2.72
3.80	6.39228	6.22217	2.66
3.85	6.49928	6.33030	2.60
3.90	6.60846	6.44059	2.54
3.95	6.71982	6.55301	2.48
4.00	6.83333	6.66756	2.43



4. Using RK4 Method (Test 2)

Inputs:

choice (method specified) = 2 (RK4 Method)

x0 (lower limit) = 0.5

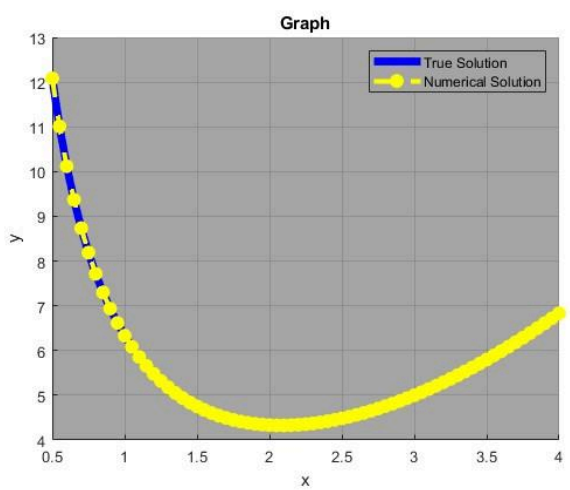
xf (upper limit) = 4

y0 (value of y at x0) = 12.08334

h (interval/step size) = 0.05

```
Choose Numerical Method to be used:
  1) Euler's Method
  2) Runge-Kutta (Order 4)
What is your choice? 2

You have chosen Runge-Kutta (Order 4)!

Enter the LOWER limit for x range (>0): 0.5

Enter the UPPER limit for x range (> lower): 4

Enter the value for y at x = 0.5: 12.08334

Enter the interval or step size (>0): 0.05

Tabulated solution:
```

x	y_true	y_RK4	|ε_t| (%)
0.50	12.08334	12.08334	0.00
0.55	11.00993	11.00993	0.00
0.60	10.12001	10.12001	0.00
0.65	9.37161	9.37161	0.00
0.70	8.73477	8.73477	0.00
0.75	8.18750	8.18750	0.00
0.80	7.71334	7.71334	0.00
0.85	7.29966	7.29966	0.00
0.90	6.93667	6.93667	0.00
0.95	6.61663	6.61663	0.00
1.00	6.33334	6.33334	0.00
1.05	6.08179	6.08179	0.00
1.10	5.85788	5.85788	0.00
1.15	5.65823	5.65823	0.00
1.20	5.48000	5.48000	0.00
1.25	5.32084	5.32084	0.00
1.30	5.17872	5.17872	0.00
1.35	5.05195	5.05195	0.00
1.40	4.93905	4.93905	0.00
1.45	4.83877	4.83877	0.00
1.50	4.75000	4.75000	0.00
1.55	4.67180	4.67180	0.00
1.60	4.60334	4.60334	0.00
1.65	4.54387	4.54387	0.00
1.70	4.49275	4.49275	0.00
1.75	4.44941	4.44941	0.00
1.80	4.41334	4.41334	0.00
1.85	4.38408	4.38408	0.00
1.90	4.36123	4.36123	0.00
1.95	4.34442	4.34442	0.00
2.00	4.33333	4.33333	0.00
2.05	4.32766	4.32766	0.00
2.10	4.32714	4.32714	0.00
2.15	4.33153	4.33153	0.00
2.20	4.34061	4.34061	0.00
2.25	4.35417	4.35417	0.00
2.30	4.37203	4.37203	0.00
2.35	4.39403	4.39403	0.00
2.40	4.42000	4.42000	0.00
2.45	4.44981	4.44981	0.00
2.50	4.48333	4.48333	0.00
2.55	4.52044	4.52044	0.00
2.60	4.56103	4.56103	0.00
2.65	4.60499	4.60499	0.00
2.70	4.65222	4.65222	0.00
2.75	4.70265	4.70265	0.00
2.80	4.75619	4.75619	0.00
2.85	4.81276	4.81276	0.00
2.90	4.87230	4.87230	0.00
2.95	4.93473	4.93473	0.00
3.00	5.00000	5.00000	0.00
3.05	5.06805	5.06805	0.00
3.10	5.13882	5.13882	0.00
3.15	5.21226	5.21226	0.00
3.20	5.28833	5.28833	0.00
3.25	5.36699	5.36699	0.00
3.30	5.44818	5.44818	0.00
3.35	5.53188	5.53188	0.00
3.40	5.61804	5.61804	0.00
3.45	5.70663	5.70663	0.00
3.50	5.79762	5.79762	0.00
3.55	5.89098	5.89098	0.00
3.60	5.98667	5.98667	0.00
3.65	6.08467	6.08467	0.00
3.70	6.18496	6.18496	0.00
3.75	6.28750	6.28750	0.00
3.80	6.39228	6.39228	0.00
3.85	6.49928	6.49928	0.00
3.90	6.60846	6.60846	0.00
3.95	6.71982	6.71982	0.00
4.00	6.83333	6.83333	0.00



5. Error Handling

```
This script solves the differential equation dy/dx + (1/x)y = x,  x > 0

Choose Numerical Method to be used:
  1) Euler's Method
  2) Runge-Kutta (Order 4)
What is your choice? 3
Invalid choice. Please enter 1 or 2.

Choose Numerical Method to be used:
  1) Euler's Method
  2) Runge-Kutta (Order 4)
What is your choice? 1

You have chosen Euler's Method!

Enter the LOWER limit for x range (>0): p
Invalid input. Please enter a numeric value.

Enter the LOWER limit for x range (>0): -9
Invalid, x0 must be finite and > 0.

Enter the LOWER limit for x range (>0): 3

Enter the UPPER limit for x range (> lower): 2
Invalid, xf must be finite and > x0.

Enter the UPPER limit for x range (> lower): o
Invalid input. Please enter a numeric value.

Enter the UPPER limit for x range (> lower): 1
Invalid, xf must be finite and > x0.

Enter the UPPER limit for x range (> lower): 7
```

```
Enter the value for y at x = 3: 5

Enter the interval or step size (>0): p
Invalid input. Please enter a numeric value.

Enter the interval or step size (>0): -1
Invalid, h must be finite and > 0.

Enter the interval or step size (>0): 1
```